Agent-to-Agent Propagation in an Agentic Mesh Architecture

A compromise of one agent can cascade through the network at machine speed.

- Normal Information Flow
- Poisoned / Malicious Flow
- Monitoring / Observability

User / Task Initiator

Goal: Market analysis and execute investment strategy

1 RESEARCH AGENT (Data Collection)

COMPROMISED

Attacker injects poisoned data or manipulated findings.

Produces poisoned research output

2 ANALYSIS AGENT (Synthesis)

Receives poisoned research and incorporates it into analysis.

Generates biased analysis

3 DECISION AGENT (Planning)

Uses biased analysis to form incorrect conclusions.

Creates harmful plan / decision

4 EXECUTION AGENT (Action)

Receives harmful plan and executes actions as instructed.

Carries out harmful actions

5 EFFECT AGENT (Impact)

Actions result in unintended, harmful outcomes.

Real-world Impact achieved

HARMFUL OUTCOME

Financial loss, data breach, reputation damage, regulatory impact, etc.

AGENTIC MESH PLATFORM

Shared Services (Identity, Memory, Tools, Communication Bus)

Used by all agents for collaboration and data exchange

HOW THE ATTACK CASCADES (AT MACHINE SPEED)

1. Attacker compromises Research Agent and injects poisoned data.
   → 20 ms
2. Poisoned information passed to Analysis Agent and accepted as valid.
   → 30 ms
3. Biased analysis passed to Decision Agent and used to form plan.
   → 25 ms
4. Harmful plan passed to Execution Agent and executed automatically.
   → 40 ms
5. Actions take effect in the real world (damage done).

TOTAL TIME ~115 ms

Attack completes before human detection is likely.

WHY THIS IS DANGEROUS

- **Operates at machine speed** — Entire chain executes in milliseconds.
- **Trust is transitive** — Each agent trusts output from the previous agent.
- **No single-agent anomaly** — Behavior of each agent may appear normal in isolation.
- **Impact is amplified** — Small initial compromise leads to large-scale harm.

KEY CONTROLS TO BREAK THE CHAIN

- **Zero Trust Between Agents** — No implicit trust. Verify every message.
- **Output Validation** — Validate data and reasoning at each hop.
- **Policy Guardrails** — Enforce allowed actions and constraints.
- **Behavior Monitoring** — Continuously monitor patterns across the mesh.
- **Automated Containment** — Detect, isolate, and revoke at the first sign of risk.